

- **STUN** Servers, which help to discover the public IP address and NAT/Firewall type if any, of a device. When a WebRTC session starts, it contacts the STUN server, which returns the device's public IP and port number. This information is used to generate ICE candidates for the peer-to-peer connection setup.
- **TURN** Servers, which are used when STUN is not sufficient to establish a peer-to-peer connection—typically such as when devices are behind symmetric NATs, which STUN cannot traverse. TURN servers act as a relay between the peers, routing the media traffic between them.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	URLs	list[string]	max 10 [max 2000]				M
1	Username	string	max 508				O
2	Credential	string	max 512				O
3	CAID	TLSCAID	0 to 65534				O

11.4.5.3.1. URLs Field

This field SHALL specify a list of URLs pointing to the STUN and/or TURN servers. The URL scheme distinguishes whether it is a STUN or TURN server (**stun:**, **stuns:**, **turn:**, or **turns:** respectively). This field maps to the **RTCIceServer urls** [<https://www.w3.org/TR/webrtc/#dom-rtciceserver-urls>] field.

11.4.5.3.2. Username Field

(Optional for STUN, usually required for TURN) The **RFC 8489** compliant UTF-8 encoded username required for authentication with the STUN or TURN servers found in the URLs field. This field maps to the **RTCIceServer username** [<https://www.w3.org/TR/webrtc/#dom-rtciceserver-username>] field.

11.4.5.3.3. Credential Field

(Optional for STUN, usually required for TURN) The **RFC 8489** compliant UTF-8 encoded short-term credential (password) used for authentication with the STUN or TURN servers found in the URLs field. This field maps to the **RTCIceServer credential** [<https://www.w3.org/TR/webrtc/#dom-rtciceserver-credential>] field.

11.4.5.3.4. CAID Field

This field represents the TLSRCAC via its assigned TLSCAID (see Chapter 14, **Certificate Authority ID (CAID) Mapping** and **TLS Certificate Management Commands** sections in **[MatterCore]**) that will validate the certificate chain presented by the entries in the urls field. It SHALL be set to a valid value if a **turns:** or **stuns:** url is present in the urls field and SHALL be used to validate those servers' presented TLS root certificates.

11.4.5.4. ICECandidateStruct Type

This type SHALL specify the **RFC 8825** compliant **ICE Candidate** used to facilitate the negotiation of peer-to-peer connections through NATs (Network Address Translators) and firewalls. It mimics the model used in the W3C WebRTC API **RTCIceCandidate** [<https://www.w3.org/TR/webrtc/#rtcicecandidate->